

Stroop Style Cue Conflict Analysis in CLIP ViT Using Zero Shot Evaluation and Layer Wise White Box Attribution

1st Aishwarya Vijay

Msc Artificial Intelligence

Queen Mary University Of London

ec25163@qmul.ac.uk

2nd Ziquan Liu

Department Name

Queen Mary University Of London

email address

Abstract—Vision-language models such as CLIP learn a shared representation between images and natural language and are commonly used for zero-shot classification by comparing an input image with a set of text prompts. However, when an image contains multiple competing cues, it is not always clear which cue the model prioritizes. This work investigates CLIP’s behavior under Stroop-style visual-semantic conflict, where the written word, the visual font color, and the background color may provide different color signals. A synthetic dataset of 20,000 images was generated with ten color classes and four controlled conditions: congruent, incongruent, neutral, and background-conflict. The correct label for every image is the visual font color. CLIP ViT-B/32 and CLIP ViT-B/16 were evaluated in a zero-shot setting using prompt-based font-color classification. The black-box evaluation showed that CLIP performs perfectly on congruent images and strongly on neutral images, but fails severely when the written word conflicts with the font color. In the ViT-B/16 experiment, CLIP predicted the written word in 99.00% of incongruent samples and 95.88% of background-conflict samples. To investigate this failure internally, gradient-based patch attribution and layer-wise transformer attribution were applied to CLIP ViT-B/16. The white-box results show that attribution becomes increasingly concentrated around the text region in middle-to-late visual transformer layers. However, image-text similarity remains consistently higher for the written-word prompt than for the correct font-color prompt. These findings suggest that CLIP’s visual-language alignment strongly prioritizes semantic word identity over visual font color under Stroop-style conflict.

Index Terms—CLIP, Vision Transformer, Stroop effect, vision-language models, interpretability, zero-shot classification, white-box analysis, gradient attribution, prompt-based evaluation

I. INTRODUCTION

Vision-language models have become an important class of multimodal models because they connect visual information with natural language. CLIP is one of the most widely used vision-language models. It consists of an image encoder and a text encoder. The image encoder maps an image into an embedding space, while the text encoder maps a text prompt into the same embedding space. CLIP then computes the similarity between the image and text embeddings [1].

Unlike a conventional classifier, CLIP does not directly output a fixed class label such as *red*, *green*, or *blue*. Instead, it compares an image with several text prompts and

assigns the image to the prompt with the highest similarity score. This makes CLIP useful for zero-shot classification because new categories can be introduced through text prompts without retraining the model.

However, this also raises an important interpretability question. When an image contains multiple possible cues, which cue does CLIP actually follow? This question becomes especially important in images containing readable text because the model may rely on semantic information from the text rather than purely visual properties. In a human Stroop task, the semantic meaning of a word can interfere with the perception or reporting of its visual color [4]. A similar type of conflict can be constructed for a vision-language model by displaying a color word in a different font color.

To study this issue, we designed a synthetic Stroop-style dataset. In a Stroop-style image, the semantic meaning of a word may conflict with its visual appearance. For example, the word *ORANGE* may be written in yellow font. In this case, the semantic word cue indicates orange, but the visual font-color cue indicates yellow. If the task is to identify the font color, the correct answer is yellow, not orange.

This work investigates whether CLIP follows the visual font color or the written word when these cues conflict. The study also includes background-color conflict, where the written word, font color, and background color may all represent different color cues. The goal is not only to measure CLIP’s accuracy, but also to understand why the model gives a particular output by inspecting internal visual transformer layers.

The main contributions of this work are as follows:

- A synthetic Stroop-style dataset with controlled word, font-color, and background-color attributes.
- A black-box zero-shot evaluation of CLIP ViT-B/32 and CLIP ViT-B/16 under color-word conflict.
- A prompt-robust analysis to identify whether errors correspond to word bias, background bias, or other errors.
- A white-box layer-wise attribution analysis of CLIP ViT-B/16 to inspect which visual regions and transformer layers support the model’s image-text similarity scores.

II. DATASET GENERATION

A. Dataset Objective

The dataset was designed to test whether a vision-language model can correctly identify the visual font color of a word image when other cues are present. Each image contains a single word displayed in a specific font color on a specific background color. For example, an image may have

word = ORANGE, font color = yellow,
background = red.

The correct label for this image is `yellow` because the task is to identify the font color. The written word and background color are treated as distractor cues.

The dataset was generated synthetically so that every important factor could be controlled. This is useful for interpretability experiments because it allows us to know exactly which cue the model followed when it made a prediction.

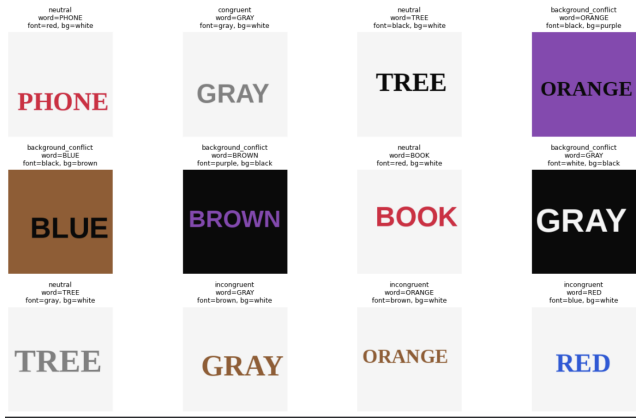


Fig. 1. Examples from the synthetic Stroop-style dataset. The correct label is always the visual font color.

B. Image Format

All images were generated at a resolution of 224×224 pixels. This resolution was selected because CLIP Vision Transformer models commonly use 224×224 image inputs. Each image contains one centered word with small random variation in font, size, and position. The purpose of this variation was to avoid making the dataset too uniform while still keeping the task controlled.

The dataset generation process included the following steps:

- 1) Select an experimental condition.
- 2) Select the written word.
- 3) Select the font color.
- 4) Select the background color.
- 5) Render the word on a 224×224 RGB canvas.
- 6) Save the generated image.
- 7) Save a binary text mask.
- 8) Store metadata such as word, font color, background color, label, and prompts.

C. Color Classes

The dataset contains ten target classes. These classes correspond to possible font colors. The same set of colors was also used for written color words and background colors. This allowed the dataset to create controlled conflict between the written word, the font color, and the background color.

TABLE I
COLOR CLASSES USED IN THE SYNTHETIC DATASET

Class Index	Color Class
1	red
2	blue
3	green
4	yellow
5	orange
6	purple
7	brown
8	gray
9	black
10	white

The label space is defined as

$$C = \{\text{red, blue, green, yellow, orange, purple, brown, gray, black, white}\}. \quad (1)$$

For every generated image, the ground-truth label is

$$y = \text{font_color}. \quad (2)$$

D. Word Vocabulary

Two word groups were used. The first group contains ten color words:

RED, BLUE, GREEN, YELLOW, ORANGE, PURPLE,
BROWN, GRAY, BLACK, WHITE.

These words create semantic conflict when the word meaning differs from the actual font color.

The second group contains ten neutral words:

CAT, DOG, SHIP, TREE, HOUSE, CAR, BOOK,
PHONE, CHAIR, MUSIC.

Neutral words do not have a color meaning. They were included to test whether CLIP can identify font color when there is no semantic color-word distractor.

E. Experimental Conditions

The dataset was divided into four conditions: congruent, incongruent, neutral, and background-conflict. Table II summarizes these conditions.

TABLE II
EXPERIMENTAL CONDITIONS

Condition	Example	Description
Congruent	RED in red on white	Word meaning and font color match
Incongruent	RED in green on white	Word meaning and font color conflict
Neutral	DOG in green on white	Word has no color meaning
Background-conflict	RED in green on yellow	Word, font color, and background conflict

The background-conflict condition is the strongest conflict condition because it contains three competing cues: written

word color, font color, and background color. For example, if RED is written in green on a yellow background, the written word suggests red, the background suggests yellow, but the correct answer is green.

F. Dataset Size

The dataset was generated with an equal number of images per condition:

$$N_{\text{per condition}} = 5000. \quad (3)$$

Since there are four conditions, the total dataset size is

$$N_{\text{total}} = 4 \times 5000 = 20000. \quad (4)$$

TABLE III
DATASET DISTRIBUTION

Condition	Number of Images
Congruent	5,000
Incongruent	5,000
Neutral	5,000
Background-conflict	5,000
Total	20,000

G. Text Mask Generation

For every generated image, a binary text mask was also saved. The text mask has the same resolution as the original image. In the mask, white pixels correspond to the text region, and black pixels correspond to the background region. The mask was not given to CLIP during prediction. It was used only during white-box analysis to measure whether attribution heatmaps were focused on the text or background.

H. Metadata

A metadata CSV file was created. Each row corresponds to one image. The main fields include image identifier, image path, mask path, word, word color, font color, background color, label, condition, and prompt strings.

TABLE IV
METADATA FIELDS

Field	Description
image_id	Unique image identifier
image_path	Path to generated image
mask_path	Path to text mask
word	Word rendered in the image
word_color	Semantic color of the word, if applicable
font_color	Actual visual font color
background_color	Background color
label	Ground-truth label, equal to font color
condition	Dataset condition
font_prompt	Prompt describing the font color
word_prompt	Prompt describing the written word
background_prompt	Prompt describing the background

An example metadata entry for a background-conflict image is:

```

word      ORANGE
word_color orange
font_color yellow
background_color red
label     yellow
condition background_conflict.
```

The corresponding prompts are the text is written in yellow, the word says orange, and the background is red.

III. CLIP ZERO-SHOT EVALUATION

A. CLIP Model

CLIP maps images and text into a shared embedding space [1]. Let x be an image and p be a text prompt. The normalized image and text embeddings are denoted by

$$f_I(x) = \frac{g_I(x)}{\|g_I(x)\|}, \quad f_T(p) = \frac{g_T(p)}{\|g_T(p)\|}, \quad (5)$$

where g_I and g_T are the image and text encoders. The CLIP similarity score is computed as

$$s(x, p) = f_I(x)^\top f_T(p). \quad (6)$$

Since embeddings are normalized, this dot product behaves as cosine similarity.

B. Difference Between Standard Classification and CLIP Classification

A normal image classifier directly predicts a class label using a learned classifier head. CLIP works differently: it compares an image with a set of text prompts and selects the prompt with the highest similarity. Therefore, to classify font color using CLIP, one prompt is created for each color class. The predicted color is

$$\hat{y} = \arg \max_{c \in C} s(x, p_c), \quad (7)$$

where p_c is a prompt such as the text is written in red.

C. CLIP Variants Evaluated

Two CLIP Vision Transformer variants were evaluated: ViT-B/32 and ViT-B/16. ViT-B/32 was used as a fast black-box baseline. ViT-B/16 was used as the main model for white-box analysis because it provides higher spatial resolution.

TABLE V
CLIP VARIANTS EVALUATED

Model	Patch Size	Patch Grid	Purpose
ViT-B/32	32×32	7×7	Fast baseline
ViT-B/16	16×16	14×14	White-box analysis

IV. PROMPT DESIGN

A. Font-Color Prompts

To evaluate font-color recognition, multiple prompt templates were used:

- the text is written in {color}
- the word is written in {color}
- the color of the letters is {color}
- the font color is {color}
- a word printed in {color} ink
- a {color} colored word
- a word in {color} text

For each image, the model compared the image with all color prompts. Similarity scores were averaged across templates. The color with the highest average similarity was selected as the prediction.

B. Word-Reading Prompts

To measure whether CLIP could read the written word, the following templates were used:

- the word says {color}
- the text reads {color}
- the written word is {color}
- the word in the image is {color}

This evaluation was applied to color-word samples only.

C. Background Prompts

To measure whether CLIP could identify background color, the following templates were used:

- the background is {color}
- the background color is {color}
- an image with a {color} background
- a word on a {color} background

V. ERROR CATEGORIZATION

Each font-color prediction was categorized into one of four error types. This categorization is important because simple accuracy only tells whether the model is correct or incorrect. The error type reveals which cue the model followed when it was wrong.

TABLE VI
ERROR TYPE DEFINITIONS

Error Type	Definition
correct_font_color	Prediction equals the true font color
word_bias	Prediction equals the written word color
background_bias	Prediction equals the background color
other_error	Prediction is neither font, word, nor background color

For example, if the word is ORANGE, the font color is yellow, the background is red, and the prediction is orange, then the prediction is categorized as `word_bias` because it matches the semantic meaning of the written word.

VI. BLACK-BOX RESULTS

A. ViT-B/32 Results

The first black-box evaluation used CLIP ViT-B/32. The overall font-color accuracy was 49.89%. Condition-wise accuracy is shown in Table VII.

TABLE VII
ViT-B/32 FONT-COLOR ACCURACY BY CONDITION

Condition	Font-Color Accuracy
Background-conflict	0.0432
Congruent	1.0000
Incongruent	0.0962
Neutral	0.8562

Since the dataset has ten color classes, random chance is $1/10 = 0.10$. Thus, the incongruent condition is approximately chance-level, and the background-conflict condition is below chance.

The prompt-robust ViT-B/32 bias table is shown in Table VIII. These results show that ViT-B/32 is strongly biased toward the written word in conflict conditions.

TABLE VIII
PROMPT-ROBUST ViT-B/32 BIAS TABLE

Condition	Background Bias	Correct Font	Other Error	Word Bias
Background-conflict	0.0636	0.0154	0.0000	0.9210
Congruent	0.0000	1.0000	0.0000	0.0000
Incongruent	0.0000	0.0242	0.0000	0.9758
Neutral	0.0642	0.8484	0.0874	0.0000

B. ViT-B/16 Results

The full 20,000-image evaluation was repeated using CLIP ViT-B/16. Each condition contained 5,000 images. The ViT-B/16 bias table is shown in Table IX.

TABLE IX
ViT-B/16 BIAS TABLE

Condition	Background Bias	Correct Font	Other Error	Word Bias
Background-conflict	0.0316	0.0096	0.0000	0.9588
Congruent	0.0000	1.0000	0.0000	0.0000
Incongruent	0.0000	0.0100	0.0000	0.9900
Neutral	0.0010	0.8704	0.1286	0.0000

C. Interpretation of Black-Box Results

The black-box results show a consistent pattern. First, CLIP performs perfectly on congruent examples. This is expected because the written word and font color match. Second, CLIP performs well on neutral examples. In the ViT-B/16 evaluation, the neutral condition achieved 87.04% correct font-color prediction. This suggests that CLIP is capable of identifying font color when the word itself does not carry a color meaning.

Third, CLIP fails severely in conflict conditions. In the incongruent condition, ViT-B/16 predicted the written word in 99.00% of cases. In the background-conflict condition, it predicted the written word in 95.88% of cases. This means that

CLIP’s failure is systematic, not random. The model strongly prioritizes semantic word identity over visual font color.

VII. WHITE-BOX TESTING

A. Purpose of White-Box Testing

Black-box testing shows what CLIP predicts, but it does not explain why the model gives that output. Therefore, white-box testing was performed to inspect internal visual evidence. The white-box analysis asks which image regions support CLIP’s image-text similarity score, how this evidence changes across visual transformer layers, and whether the model internally aligns the image more strongly with the word prompt or the font-color prompt.

B. Patch-Level Attribution

The first white-box method was patch-level gradient attribution. The CLIP ViT-B/16 patch embedding layer was used because it produces a 14×14 grid of visual patches. For each image-prompt pair, the image and prompt were passed through CLIP, the image-text similarity score was computed, the score was backpropagated through the visual encoder, and gradients and activations at the patch embedding layer were used to create a heatmap. The heatmap was then resized to the image size and overlaid on the original image.

C. Layer-Wise Transformer Attribution

Patch-level attribution gives a general heatmap, but it does not show how evidence changes across the transformer. Therefore, layer-wise attribution was computed at selected visual transformer blocks: layers 0, 3, 6, 9, and 11. The final analysis focused mainly on layers 0, 3, 6, and 9. Layer 11 was excluded from the main interpretation because the current gradient activation method often produced near-zero patch relevance at the final block. This may occur because information has already been compressed into the global image representation.

For a selected layer l , patch relevance was computed using gradient multiplied by activation:

$$R_i^{(l)} = \left| A_i^{(l)} \odot \frac{\partial s}{\partial A_i^{(l)}} \right|, \quad (8)$$

where $A_i^{(l)}$ is the activation of patch i at layer l , s is the CLIP image-text similarity score, and $R_i^{(l)}$ is the relevance of patch i at layer l . The class token was removed, and the remaining patch token relevance values were reshaped into a 14×14 patch grid.

D. Text Enrichment Metric

To quantify whether a heatmap focused on the text region, the binary text mask was used. The following metrics were computed:

$$E_{text} = \frac{\sum_{i \in M_{text}} H_i}{\sum_i H_i}, \quad (9)$$

$$E_{bg} = \frac{\sum_{i \in M_{bg}} H_i}{\sum_i H_i}, \quad (10)$$

where H_i is the heatmap value at pixel i , M_{text} is the text mask, and M_{bg} is the background mask. The text area share is

$$A_{text} = \frac{|M_{text}|}{|M_{image}|}. \quad (11)$$

The text enrichment score is defined as

$$\text{Text Enrichment} = \frac{E_{text}}{A_{text}}. \quad (12)$$

If the text enrichment score is greater than 1, then the heatmap is more concentrated on the text region than expected by area alone.

VIII. WHITE-BOX RESULTS

A. Representative Word-Bias Example

A representative background-conflict word-bias example was selected. The word was ORANGE, the font color was yellow, and the background color was red. The correct answer should be yellow. However, CLIP predicted orange, which matches the written word.

B. Prompt Similarity for Representative Example

For the selected example, three prompts were compared. The word prompt received the highest similarity score, as shown in Table X. This explains the black-box output: CLIP predicts orange because the image representation aligns more strongly with the prompt the word says orange than with the correct font-color prompt.

TABLE X
PROMPT SIMILARITY FOR REPRESENTATIVE WORD-BIAS EXAMPLE

Prompt Type	Prompt	Similarity
Font prompt	the text is written in yellow	0.2494
Word prompt	the word says orange	0.3232
Background prompt	the background is red	0.2593

C. Layer-Wise Attribution for Word Prompt

For the word prompt the word says orange, text enrichment increased from early to deeper visual transformer layers. The values are shown in Table XI.

TABLE XI
SINGLE EXAMPLE TEXT ENRICHMENT FOR WORD PROMPT

Layer	Text Enrichment
0	1.9453
3	2.6473
6	4.8321
9	5.9986

D. Layer-Wise Attribution for Font Prompt

For the correct font-color prompt the text is written in yellow, the text enrichment values also increased with depth, as shown in Table XII. The font prompt also activates the text region because font color exists on the letters. However, the CLIP similarity score for the font prompt is lower than the word prompt.

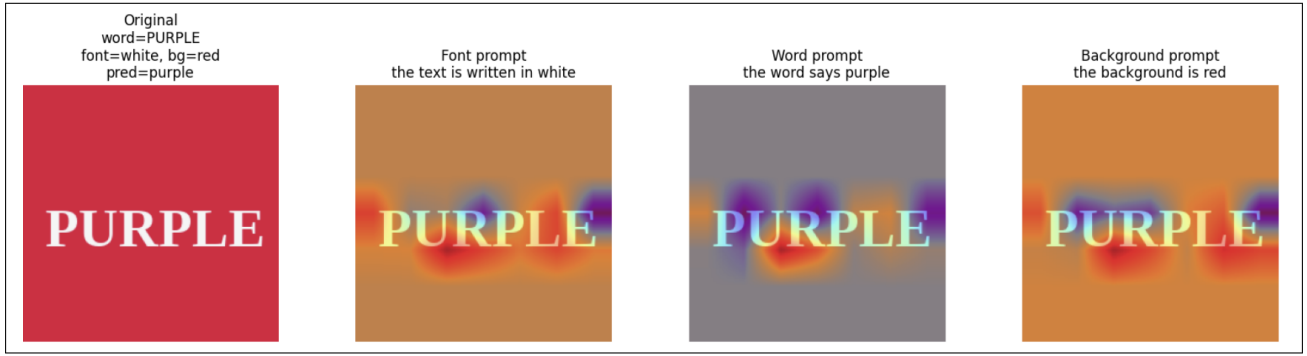


Fig. 2. Patch-level ViT-B/16 attribution for font, word, and background prompts. The CAM shape is 14×14 .

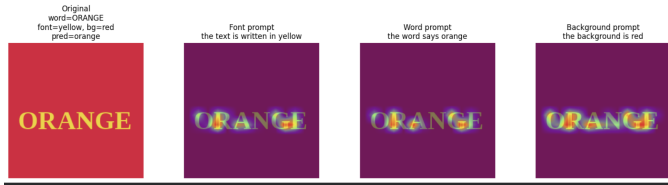


Fig. 3. Representative word-bias example. The correct label is yellow, but CLIP predicts orange.

TABLE XII
SINGLE EXAMPLE TEXT ENRICHMENT FOR FONT PROMPT

Layer	Text Enrichment
0	2.0097
3	3.1623
6	4.2853
9	5.1064

E. Interpretation of Single Example

The representative example shows an important behavior. Both the font prompt and the word prompt produce attribution on the text region. This means CLIP is not ignoring the text area. However, the similarity score is higher for the word prompt. Therefore, the issue is not that CLIP fails to look at the correct region. Instead, the issue is that the representation of the text region aligns more strongly with semantic word identity than with visual font color. In other words, CLIP looks at the text region, but interprets it more strongly as a word than as a colored visual object.

IX. AGGREGATE WHITE-BOX ANALYSIS

A. Groups Analyzed

To avoid relying on a single example, layer-wise attribution was repeated across multiple examples from the following groups: background_conflict__word_bias, background_conflict__correct_font_color, incongruent__word_bias, and neutral__correct_font_color. The most important groups for Stroop conflict are background_conflict__word_bias and

incongruent__word_bias because these contain images where CLIP incorrectly follows the written word.

B. Mean Text Enrichment

The aggregate text enrichment results are shown in Table XIII. In the background-conflict word-bias group, both prompt types become increasingly text-focused across layers. The word prompt becomes stronger than the font prompt at layer 6, and both are highly text-focused at layer 9.

C. Mean CLIP Similarity

The aggregate CLIP similarity results are shown in Table XIV. For the background-conflict word-bias group, the mean similarity of the word prompt is 0.3330, while the mean similarity of the font prompt is 0.2468. For the incongruent word-bias group, the word prompt similarity is 0.3276, while the font prompt similarity is 0.2430. These results confirm that CLIP’s image representation aligns more strongly with the written-word prompt than with the font-color prompt in word-bias cases.

D. Layer-Wise Trend

For background-conflict word-bias examples, the text enrichment trend is summarized in Table XV. The plot in Fig. ?? should be inserted after this table.

X. DISCUSSION

The results provide both behavioral and internal evidence of CLIP’s word bias. The black-box evaluation showed that CLIP ViT-B/16 predicts the written word in most conflict cases. In the incongruent condition, 99.00% of predictions were word-bias errors. In the background-conflict condition, 95.88% of predictions were word-bias errors. This indicates that CLIP is not randomly failing; it systematically prioritizes the semantic word cue.

The white-box evaluation explains this behavior. Attribution maps show that the model’s evidence is concentrated around the text region. This is expected because both the written word and font color are located in the same region. However, the word prompt receives higher CLIP similarity than the font prompt. Therefore, CLIP’s representation of the text region is more strongly aligned with word identity than font color.

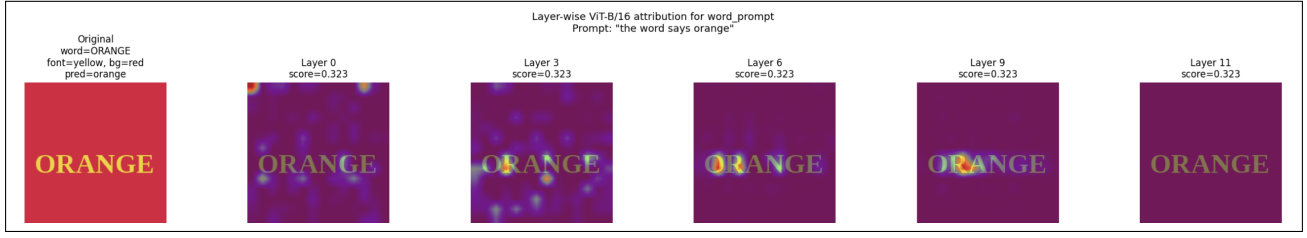


Fig. 4. Layer-wise ViT-B/16 attribution for the word prompt. The attribution becomes increasingly concentrated around the text region in deeper layers.

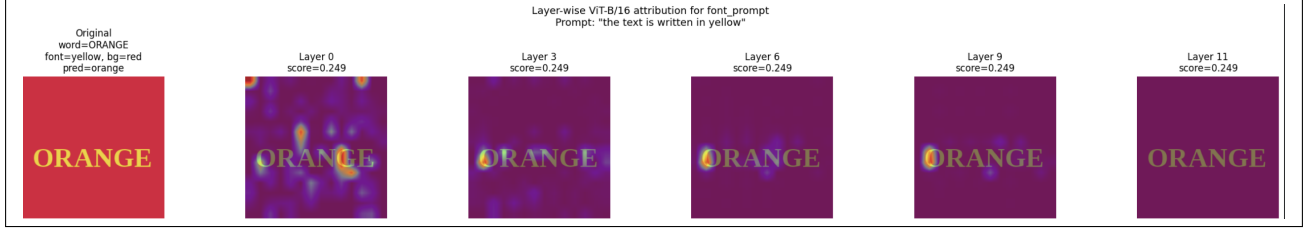


Fig. 5. Layer-wise ViT-B/16 attribution for the font prompt. The model also uses the text region for font-color evidence, but the final similarity score is lower than the word prompt.

TABLE XIII
MEAN TEXT ENRICHMENT BY GROUP, LAYER, AND PROMPT TYPE

Group	Layer	Font Prompt	Word Prompt
background_conflict__correct_font_color	0	2.8484	2.4855
background_conflict__correct_font_color	3	3.0081	3.3167
background_conflict__correct_font_color	6	4.2856	3.9543
background_conflict__correct_font_color	9	4.7526	4.9048
background_conflict__word_bias	0	1.9426	1.8481
background_conflict__word_bias	3	2.7693	2.8727
background_conflict__word_bias	6	3.7701	4.2183
background_conflict__word_bias	9	5.0756	5.0683
incongruent__word_bias	0	2.4021	2.0457
incongruent__word_bias	3	3.1920	2.7987
incongruent__word_bias	6	2.9958	3.4267
incongruent__word_bias	9	3.6949	4.3175
neutral__correct_font_color	0	2.4225	2.2024
neutral__correct_font_color	3	2.5071	3.0115
neutral__correct_font_color	6	3.0480	4.1875
neutral__correct_font_color	9	4.1290	4.6647

TABLE XIV
MEAN CLIP SIMILARITY BY GROUP AND PROMPT TYPE

Group	Font Prompt	Word Prompt
background_conflict__correct_font_color	0.2980	0.3240
background_conflict__word_bias	0.2468	0.3330
incongruent__word_bias	0.2430	0.3276
neutral__correct_font_color	0.2959	0.3420

TABLE XV
LAYER-WISE TEXT ENRICHMENT FOR BACKGROUND-CONFLICT
WORD-BIAS EXAMPLES

Layer	Font Prompt	Word Prompt
0	1.9426	1.8481
3	2.7693	2.8727
6	3.7701	4.2183
9	5.0756	5.0683

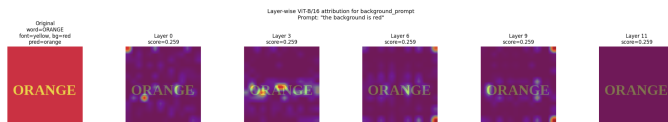


Fig. 6. Layer-wise text enrichment for background-conflict word-bias examples. Text-region relevance increases from early to later transformer layers.

This is an important distinction. The model is not simply looking at the wrong part of the image. It is looking at the correct region but extracting the semantic word cue more strongly than the visual color cue. The neutral condition supports this interpretation. When the word does not carry a color meaning, CLIP can identify font color with relatively

high accuracy. This means CLIP is capable of visual color recognition, but semantic word cues dominate when they conflict with font color.

XI. LIMITATIONS

This study has several limitations. First, the dataset is synthetic. Synthetic data allows strong control over variables, but it may not fully represent natural images containing text. Second, the white-box method used here is a custom gradient-based patch attribution approach. It is inspired by Grad-CAM-style and Grad-ECLIP-style analysis, but it is not the official Grad-ECLIP implementation. Third, the current white-box analysis focuses mainly on CLIP’s image encoder. Future work should also analyze the text encoder and text-token importance. Fourth, the analysis was performed on CLIP ViT-B/32 and ViT-B/16. Other CLIP variants, OpenCLIP models, BLIP models, or larger vision-language models may behave differently. Fifth, the prompt design uses handcrafted prompt templates. Although prompt robustness was considered, CLIP can still be sensitive to wording.

XII. CONCLUSION

This work investigated CLIP’s behavior under Stroop-style cue conflict. A synthetic dataset of 20,000 images was generated with ten color classes and four conditions. The correct label was always the visual font color.

Black-box evaluation showed that CLIP performs well when no color-word conflict exists, but fails severely when the written word conflicts with the font color. In CLIP ViT-B/16, 99.00% of incongruent predictions and 95.88% of background-conflict predictions followed the written word instead of the font color.

White-box attribution showed that visual evidence becomes increasingly concentrated around the text region in middle-to-late transformer layers. However, the image-text similarity score is consistently higher for the written-word prompt than for the correct font-color prompt. This suggests that CLIP’s internal representation of the text region is more strongly associated with semantic word identity than with visual font color.

The main conclusion is that CLIP can recognize font color when no semantic conflict exists, but under Stroop-style conflict it strongly prioritizes the written word over the visual font color.

XIII. FUTURE WORK

This paper is currently presented as a first draft, and the following directions are still ongoing parts of the work. Future work can extend this study in several ways. First, the official Grad-ECLIP implementation can be applied to compare with the custom gradient-based patch attribution used in this work [3]. Second, text-encoder attribution can be added to identify which prompt tokens contribute most to the image-text similarity score. Third, more models can be tested, including ViT-L/14, OpenCLIP, BLIP, and other multimodal models. Fourth, the synthetic dataset can be expanded with additional

visual variation, including more fonts, rotations, blur, noise, shadows, and natural image backgrounds. Fifth, CLIP can be fine-tuned on conflict-aware Stroop data to test whether word bias can be reduced. Finally, a comparison can be made between Vision Transformers, CLIP, and other vision-language models to study how different architectures resolve visual-semantic conflict.

ACKNOWLEDGMENT

The author thanks the project supervisor and reviewers for their guidance and feedback. This section can be modified or removed depending on submission requirements.

REFERENCES

- [1] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever, “Learning transferable visual models from natural language supervision,” in *Proc. Int. Conf. Mach. Learn.*, 2021, pp. 8748–8763.
- [2] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby, “An image is worth 16x16 words: Transformers for image recognition at scale,” in *Proc. Int. Conf. Learn. Represent.*, 2021.
- [3] C. Zhao, K. Wang, J. H. Hsiao, and A. B. Chan, “Grad-ECLIP: Gradient-based visual and textual explanations for CLIP,” arXiv:2502.18816, 2025.
- [4] J. R. Stroop, “Studies of interference in serial verbal reactions,” *J. Exp. Psychol.*, vol. 18, no. 6, pp. 643–662, 1935.
- [5] K. Zhou, J. Yang, C. C. Loy, and Z. Liu, “Learning to prompt for vision-language models,” *Int. J. Comput. Vis.*, vol. 130, pp. 2337–2348, 2022.
- [6] K. Zhou, J. Yang, C. C. Loy, and Z. Liu, “Conditional prompt learning for vision-language models,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 16816–16825.
- [7] M. U. Khattak, H. A. Rasheed, M. Maaz, S. Khan, and F. S. Khan, “Self-regulating prompts: Foundational model adaptation without forgetting,” in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2023, pp. 15190–15200.
- [8] T. Oikarinen and T.-W. Weng, “CLIP-Dissect: Automatic description of neuron representations in deep vision networks,” in *Proc. Int. Conf. Learn. Represent.*, 2023.
- [9] A. Madasu, Y. Gandselman, V. Lal, and P. Howard, “Quantifying and enabling the interpretability of CLIP-like models,” arXiv:2409.06579, 2024.
- [10] Y. Tang, Y. Yamada, Y. Zhang, and I. Yildirim, “When are lemons purple? The concept association bias of vision-language models,” in *Proc. Conf. Empirical Methods Natural Lang. Process.*, 2023, pp. 14333–14348.
- [11] D. Koishigarina, A. Uselis, and S. J. Oh, “CLIP behaves like a bag-of-words model cross-modally but not uni-modally,” arXiv:2502.03566, 2025.
- [12] A. M. Samin, M. F. Ahmed, and M. M. S. Rafee, “ColorFoil: Investigating color blindness in large vision and language models,” in *Proc. NAACL Student Research Workshop*, 2025.
- [13] G. Arias, R. Baldrich, and M. Vanrell, “Color in visual-language models: CLIP deficiencies,” arXiv:2502.04470, 2025.
- [14] N. Teker, R. Xiao, Z. Akata, and S. Wu, “What is the color of RED? Vision-language models prefer to read rather than see,” under review, 2026.